

FEE DEFAULTER PREDICTION SYSTEM

Predicting tuition-payment risk with explainable ML and an AI-generated intervention advisory

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Flexisaf Generative AI & Data Science Capstone

Problem Statement

School administrators need to identify students at risk of falling behind on fee payments ***before it happens***, so interventions — payment plans, financial counseling, scholarship referral — can happen proactively instead of reactively.

11.9%

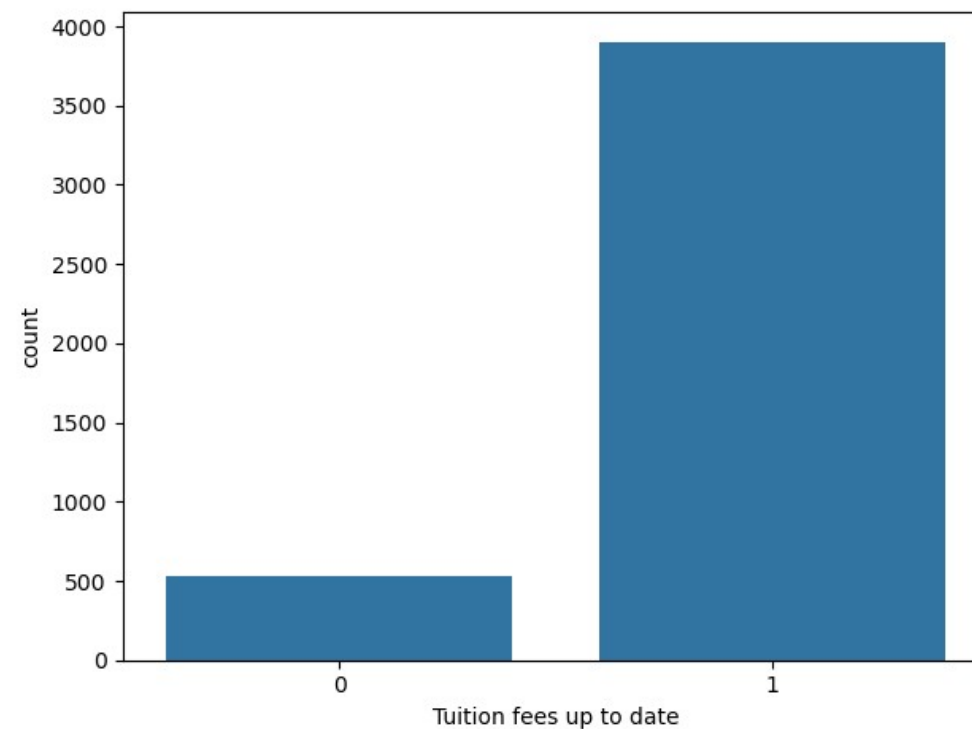
of students in the dataset are not
current on tuition

86.6%

of behind-on-fees students in this
dataset eventually dropped out

Dataset

- **4,424** real students, Portuguese higher education institution
- **37** original columns — zero missing values, zero duplicates
- **CC BY 4.0** Realinho et al. (2021), UCI ML Repository, DOI 10.24432/C5MC89
- **Reframed target** 'Tuition fees up to date' inverted → fee-default proxy (not the dataset's original dropout label)



Class balance: fee-current vs. behind-on-fees students

Methods

1

Feature Engineering

Approval-rate ratios (1st/2nd sem), zero-enrollment flags, grade trend — derived features outrank the raw columns they came from.

2

Models Compared

Decision Tree (interpretable baseline) vs. Random Forest (ensemble), both class-weighted for the ~88/12 imbalance.

3

Evaluation Priority

F1 and recall on the defaulter class over raw accuracy — a missed defaulter costs more than a false alarm.

4

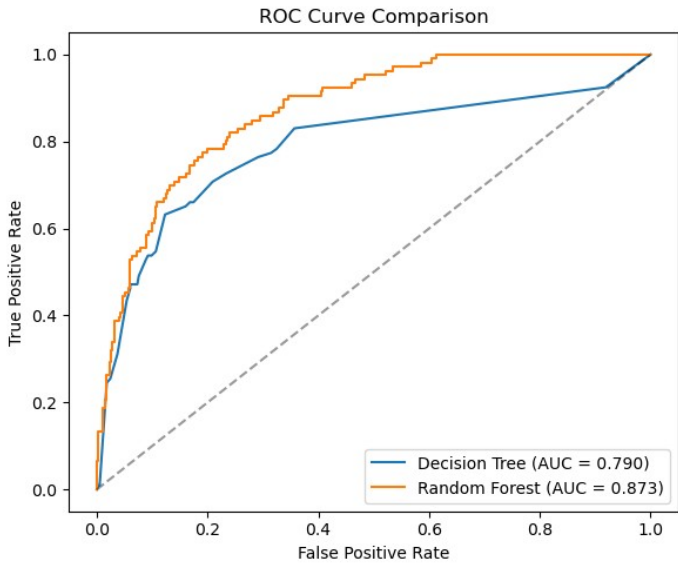
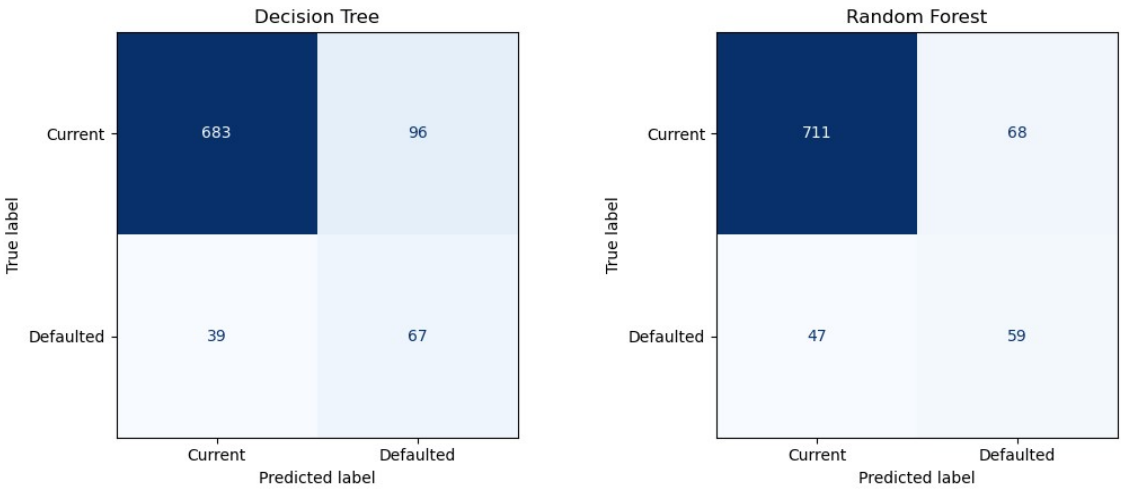
Explainability

Global feature importance + per-student SHAP values, not just a black-box score.

Model Results

Model	Threshold	Precision	Recall	F1
Decision Tree	0.5	0.411	0.632	0.498
Random Forest	0.5	0.465	0.557	0.506
Random Forest (tuned)	0.456	0.454	0.651	0.535

Tuned Random Forest beats both alternatives on every metric simultaneously — the deployed model.

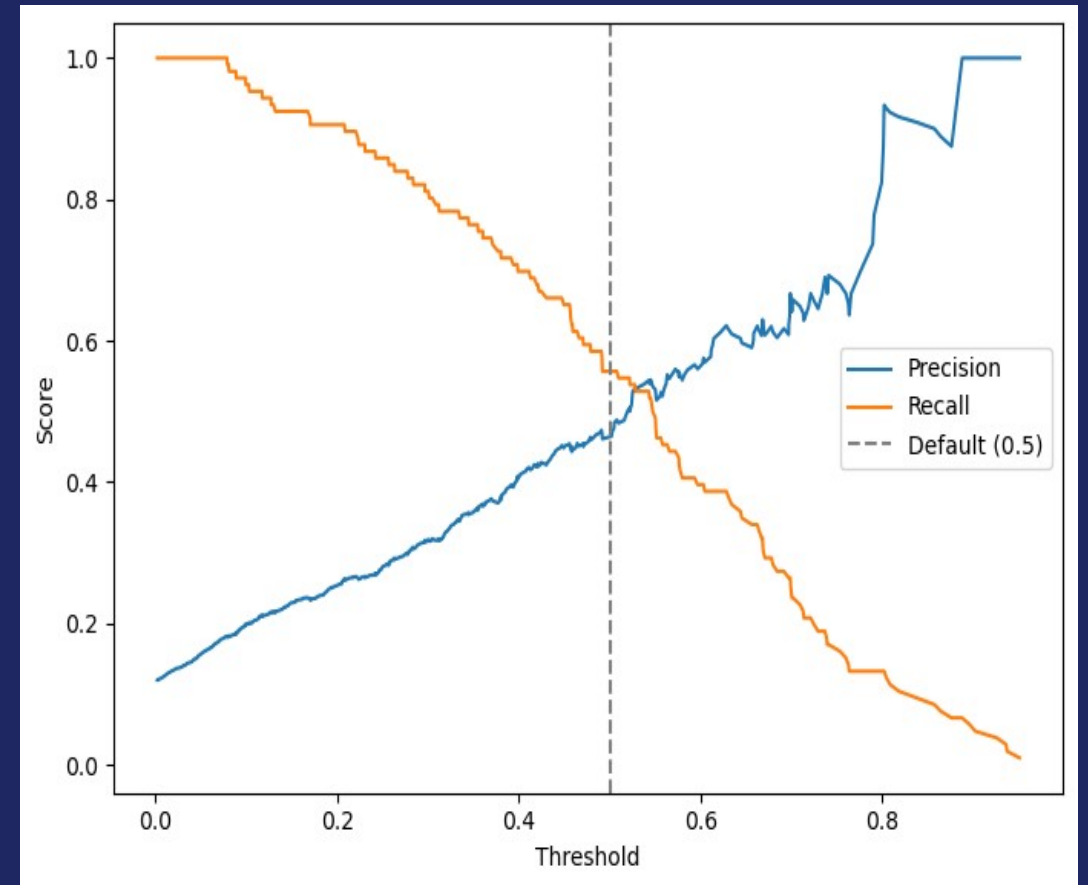


A Real Finding, Not a Formality

Threshold Tuning

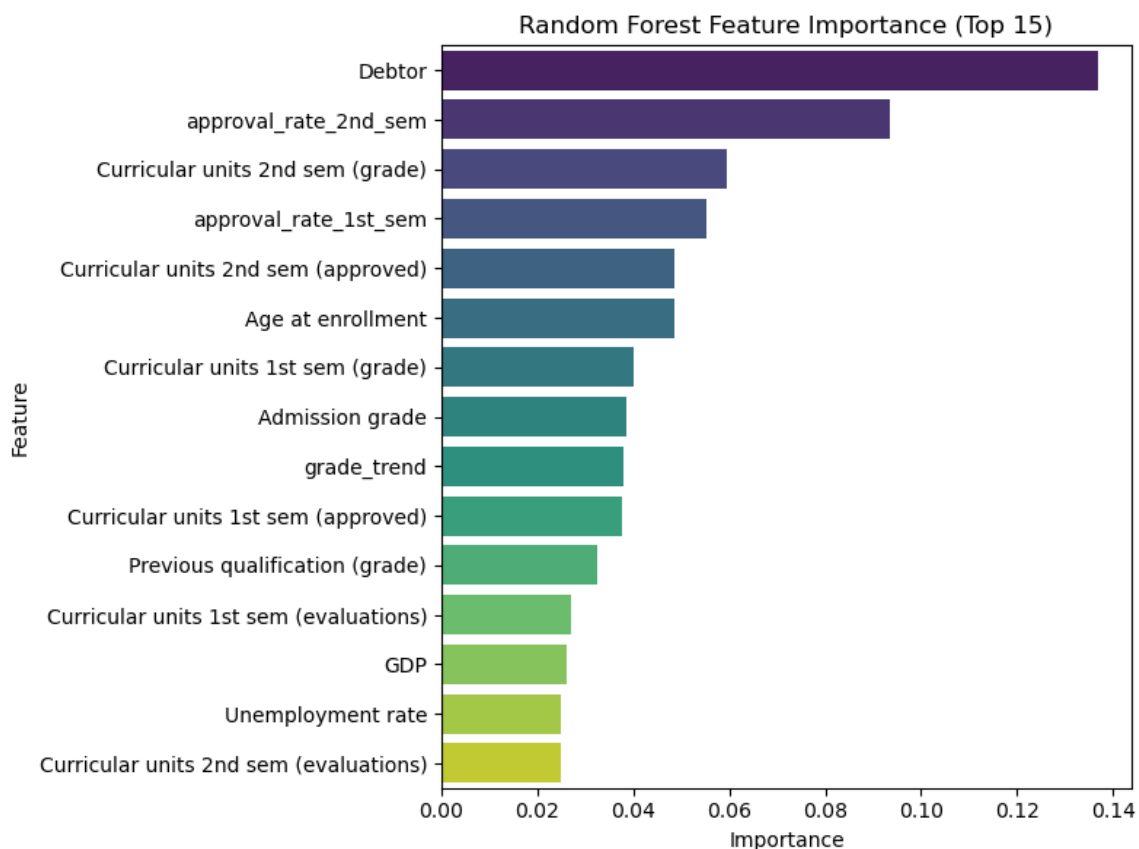
At the default 0.5 threshold, Random Forest had **lower recall than the Decision Tree** (0.557 vs 0.632) — the opposite of what “the stronger model” would suggest.

Rather than accept that, the decision threshold was tuned down to **0.456** — trading a little precision for meaningfully higher recall, beating both alternatives on every metric at once.

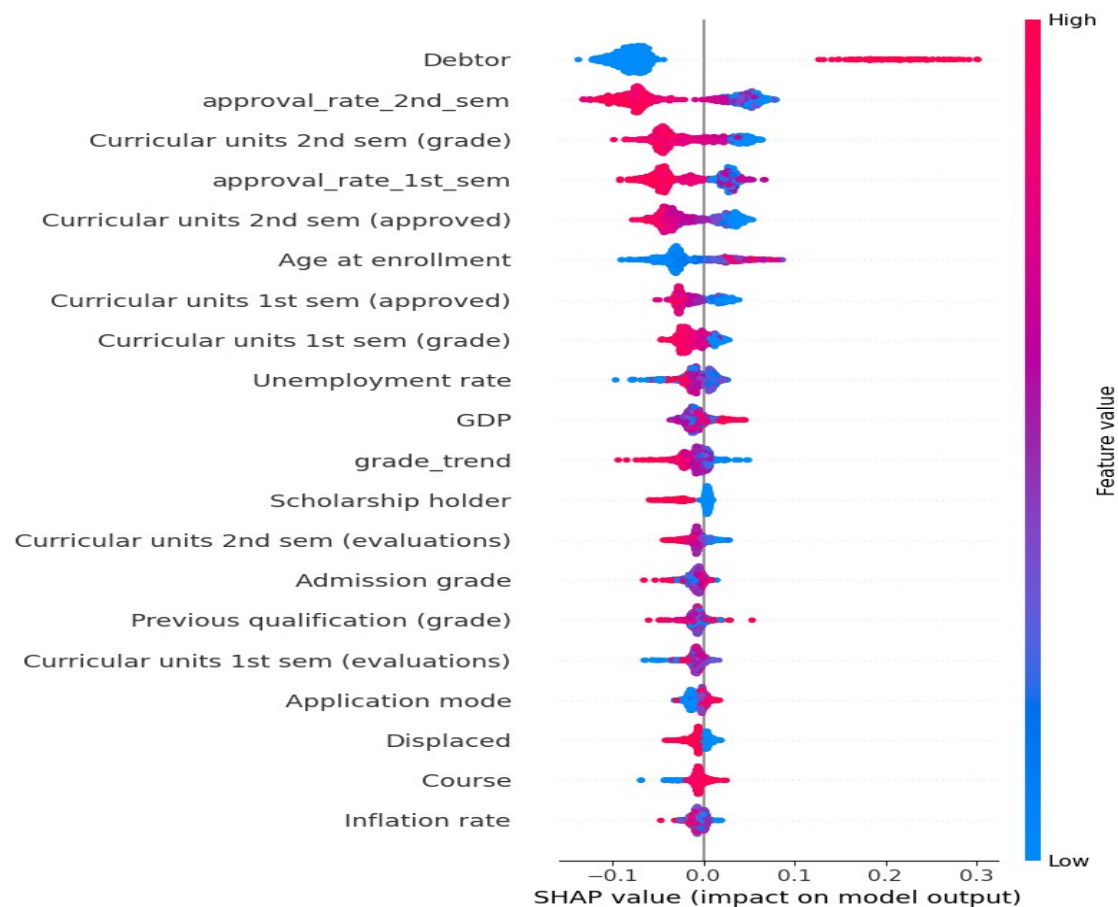


Explainability

Global feature importance shows what matters overall; SHAP shows why one student's prediction came out the way it did.



Debtor dominates both charts — confirming a pattern first spotted in manual EDA, not just a model artifact.



AI-Powered Feature

Personalized fee-recovery suggestions — SHAP factors feed an LLM (Groq) that drafts a specific admin recommendation, with a rule-based fallback.

Three real bugs found and fixed while building this:

01

Backwards direction + invented scale

Problem: Prompt passed only direction labels, not values. LLM guessed wrong and invented a 4.0 GPA scale that doesn't exist in this 0–20 dataset.

Fix: Pass real values + state the grading scale explicitly.

02

Undecoded categorical codes

Problem: Course = 33 shown raw. LLM interpreted it as a “course load of 33,” not the program code it actually is.

Fix: Decode categoricals to real labels before display or prompting.

03

Invented causal narrative

Problem: Even with correct values, the LLM added unsupported claims (e.g. a course “indicates engagement”) from statistical association alone.

Fix: Explicitly instruct the model not to speculate about why — state values and action only.

Conclusion

- Tuned Random Forest (threshold 0.456) deployed — beats both alternatives on every metric for the defaulter class.
- Debtor status, semester approval rates, and admission grade are the strongest predictors.
- The GenAI layer required real debugging — grounding values, decoding categoricals, and constraining speculation were all necessary for trustworthy output.
- Methodology transfers to real Nigerian fee data; the specific learned patterns here are population-specific (Portuguese higher education).

GitHub: github.com/Schwoz17/flexisaf-capstone

Live Demo: flexisaf-capstone.streamlit.app